

STEVEN MAN

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EDUCATION

Carnegie Mellon University

Ph.D. in Robotics (QPA: 4.0/4.0)

August 2023 – December 2028 (Expected)

Pittsburgh, PA

- **Advisors:** Dr. Aaron Johnson and Dr. Sarah Bergbreiter
- **Fellowships:** National Science Foundation Graduate Research Fellow (2025), Tau Beta Pi Fellow (2024)

University of California Santa Barbara

B.S. in Mechanical Engineering (GPA: 3.97/4.0, High Honors)

August 2019 – June 2023

Santa Barbara, CA

GRADUATE RESEARCH EXPERIENCE

Meta Reality Lab

Research Scientist Intern

August 2026 – December 2026

Redmond, WA

- Learning-based dexterous manipulation policies for high precision tasks.

Carnegie Mellon University, Robotics Institute

Graduate Researcher

August 2023 – Present

Pittsburgh, PA

- Experienced with physics simulators (MuJoCo), reinforcement learning and controls algorithms, and real-world deployment on manipulation and locomotion systems.
- Using parameter identification in physics simulation as a modeling and measurement tool to study walking dynamics of cm-scale magnetically-actuated quadrupedal robots. [1]
- Applying learned safety filters to high-dimensional control systems using latent-space dynamics. [2]
- Building a magnetic wall-climbing robot designed to maneuver on ferrous structures for industrial inspection.
- Mentoring 6 undergraduate researchers on building various scaled biped robots and developing control strategies for stable, efficient walking over uneven terrains. [4]

UNDERGRADUATE RESEARCH EXPERIENCE

Carnegie Mellon University, Robotics Institute Summer Scholars

Undergraduate Researcher

June 2022 – August 2022

Pittsburgh, PA

- Created the first micro-scale Delta robot using Two-Photon Polymerization to study 3D-printing applications in micro-robots, achieving less than 5 micron trajectory tracking precision and accuracy error.
- Modeled micro-robot motion using Simscape Multibody and analyzed its trajectory using TEMA software.
- First-authored paper published in *Science Robotics*. [3]

Hawkes Lab, UCSB

Undergraduate Researcher

January 2021 – June 2023

Santa Barbara, CA

- Built robots to mimic embryo cells to study how the robot cluster can collectively change shapes. [8]
- Designed Printed Circuit Boards and 3D-Printed assemblies to rapidly prototype embryo-inspired robots.
- Developed Arduino algorithms for light sensing and motor actuation using a photo-diode network.

Pennathur Lab, UCSB

Undergraduate Researcher

January 2022 – June 2023

Santa Barbara, CA

- Characterized a Liquid-Metal Microfluidic Energy Transducer to maximize electrical energy harvesting.
- Fabricated microfluidic devices using Deep Reactive Ion Etcher, Electron-Beam Metal Deposition, Atomic Layer Deposition, and Photolithography at UCSB's Nanofabrication facility.

INDUSTRY EXPERIENCE

SpaceX

June 2023 – August 2023

Space Lasers Mechanical Engineering Intern

Redmond, WA

- Designed, assembled, and tested a robotic optical alignment production stand to manufacture flight-level satellite hardware in 8 weeks.
- Implemented and optimized a robot automation procedure to reduce takt time by 50% compared to previous generation stands while achieving micron-precision alignment.

Johns Hopkins University Applied Physics Laboratory

June 2021 – September 2021

Mechanical Engineering Intern

Laurel, MD

- Built a mechatronic assembly for the NASA Dragonfly Mission Thermal Trim Device Tests to study the system's capacity to exchange heat with Titan's cryogenic atmosphere.
- Performed thermal balance tests on the device in a thermal chamber and analyzed its performance.

PROJECTS

Apical Robotics Capstone | UCSB

September 2022 – June 2023

- Developed the largest soft robot ever (100 ft long) for industrial pipe inspection applications and performed an onsite demonstration with the team for Bechtel Corp. [5]
- Led the design and testing of a blower fan module capable of high-volume (60 cfm) rapid flow reversal.
- Designed and built an electronics and communication system to supply the power required for pressurized robot locomotion, transmit live video feed, and perform feedback control on pressure and growth speed.
- Winner of UCSB Mechanical Engineering Capstone Innovation Award and New Venture Competition Tech Push.

Vine Endoscopy Robot | UCSB

March 2022 – June 2022

- Designed a soft vine robot for medical endoscopic applications utilizing highly-precise fabrication techniques. [7]
- Constructed a compact RF motor control circuit to allow for both extension and retraction of robot tip.
- Rapidly prototyped vine robot tip mounts using 3D-printing and reduced growing friction by 50%.

Ionic Skies Capstone | UCSB

June 2020 – June 2021

- Designed a gravity-driven launcher with a pulley mechanism to launch an ionic wind aircraft.
- Used Python to numerically model the relationship between take-off velocity and launch mass.

PUBLICATIONS & PRESENTATIONS

[1] **S. Man***, Bei Luo*, Naomi Oke, Aaron M. Johnson, Sarah Bergbreiter. "Tiny Legs or Tiny Wheels? Comparing Locomotion Morphologies for 100 mg Quadrupeds Using Real-To-Sim System Identification," *International Conference on Intelligent Robots and Systems*, 2026.

[2] K. Nakamura, A. Bishop, **S. Man**, A. M. Johnson, Z. Manchester, A. Bajcsy. "How to Train Your Latent Control Barrier Function: Smooth Safety Filtering Under Hard-to-Model Constraints," *Learning for Dynamics & Control Conference*, 2026. (Oral Presentation)

[3] **S. Man***, S. Kim*, S. Bergbreiter. "The microDelta: downscaling microscale Delta robot enables ultra-fast and high precision movement," *Science Robotics*, 2025.

[4] **S. Man***, S. Narita*, J. Macera*, N. Oke, A. M. Johnson, and S. Bergbreiter, "Zippy: The smallest power-autonomous bipedal robot," *IEEE International Conference on Robotics and Automation*, 2025.

[5] W. E. Heap, **S. Man**, V. Bassari, S. Nguyen, E. B. Yao, N. A. Tripathi, N. D. Naclerio, and E. W. Hawkes, "Large-scale vine robots for industrial inspection," *IEEE Robotics & Automation Magazine*, 2024.

[6] **S. Man**, S. Bergbreiter, and A. M. Johnson. "Hip-Leg Coordinated Quasi-Passive Biped Walker for Energy Efficiency and Step Ascension," *Dynamic Walking Conference*, 2024 (Poster Presentation)

[7] W. Heap, **S. Man**. "Novel Miniaturized Apical Extension Robot For Small Bowel Endoscopy," *UCSB ME153 Mechanical Engineering Design Fair*, 2022 (Poster Presentation)

[8] **S. Man**, M. Devlin, E. Hawkes. "Embryo Cell Robot Dynamics," *UCSB URCA Poster Colloquium*, 2022 (Poster Presentation)

GRANTS & AWARDS

National Science Foundation Graduate Research Fellowship [2025]

Tau Beta Pi Fellowship [2024]

UCSB Mechanical Engineering Capstone Innovation Award (\$1500) [2023]

UCSB New Venture Competition Tech Push 3rd Place + People's Choice (\$4500) [2023]

UCSB Edison STEM Research Scholarship (\$2000) [2022]

UCSB Gene & Susan Lucas Research Fund (\$1000) [2021, 2022]

UCSB URCA Research Grant (\$750) [2021]

LA Hacks 2021, 1st Place (\$500 Hardware) [2021]

SB Hacks VII, 2nd Place (\$300 Hardware) [2021]